DBA5103 • LECTURE SUMMARY CHEAT SHEET

SIDE 1 / 2 · WHAT EACH LECTURE COVERED (L1–L8)

L1 · INTRODUCTION: WHAT OR IS

In one line: pick decisions to maximize/minimize an objective subject to constraints, then choose the right tool to solve the model.

Covered: course road map LP → network flow → IP → DP. Software: Python + Gurobi, Excel Solver. Application areas: manufacturing, transport, airline/hotel/retail. Cases: **Ford flexible manufacturing** (dedicated vs fully flexible plants; evaluate a design with LP, max flow or simulation); **Uber/Grab** dispatch (greedy matching vs assignment/shortest path), surge pricing (supply–demand model: LP/linear algebra if non-profit, nonlinear if profit-maximizing), **UberPOOL** as a TSP with pickup before drop-off (LP, IP, heuristics); **Rue La La** price optimization (IP/LP); markdown/dynamic pricing (LP, DP).

Numbers to know: capacities (100,100,100), demands (20,140,140): dedicated sells 220, full flexibility 300. Dispatch times [2,3;10,20]: greedy 2+20=22 min, optimal 3+10=13.

Quiz angle: say which model type fits a story (LP / network / IP / DP) and why.

Watch out: greedy, one-step-at-a-time choices can be far from optimal.

L2 · LINEAR PROGRAMMING & REFORMULATION

In one line: write a word problem as a linear model, then rewrite it in a standard form that theory and solvers use.

Covered: Wyndor Glass prototype: max $3x_1+5x_2$; $x_1\le4$, $2x_2\le12$, $3x_1+2x_2\le18$, $x\ge0$; solved by the graphical/contour method, optimum (2,6), z=36. Linear algebra review: vectors, matrices, transpose, products, identity, inverse, linear combinations, when A is invertible. General LP with ≥, ≤, = rows → standard form. Tricks: free variables, absolute values, min–max. Regression with L1 / L∞ / L2 error and outlier sensitivity. Sparse recovery (compressed sensing, image reconstruction): min $\sum|x_j|$ s.t. Ax=b.

Formulas: min c^Tx, Ax=b, x≥0. ≤ row: +slack; ≥ row: −surplus. Free $x=x^+-x^-$. $|r|\le t \Leftrightarrow -t\le r\le t$. L1 and L∞ fits are LPs; L2 is quadratic.

Quiz angle: formulate from words; convert to standard form; linearize an absolute-value objective.

Watch out: "linear" means linear in the decision variables (no xy, x^2). The $t\ge\pm r$ trick needs the objective to push t down.

L3 · GEOMETRY & THE SIMPLEX METHOD

In one line: an LP optimum sits at a corner (BFS); simplex moves from corner to a better adjacent corner until no move helps.

Covered: Wyndor in Excel Solver and Gurobi; keep model and data separate. 2-D geometry: feasible region = polyhedron, extreme points, four outcomes (unique, multiple, unbounded, infeasible). Binding constraints and slack. BFS in n dimensions = n linearly independent active constraints. Basic solutions in standard form: pick m columns, set the rest to 0, solve, check ≥0. Degeneracy. Convex sets/functions (local optimum = global). Adjacent BFSs. Simplex: find a BFS → reduced costs → entering variable → ratio test → pivot.

Formulas: $x_B=B^{-1}b$; $r_j=c_j-c_BB^{-1}A_j$. MIN optimal when all $r_j\ge0$. Example max x_1+x_2; $x_1+2x_2\le3$, $2x_1+x_2\le3$ → (1,1), z=2.

Quiz angle: list basic solutions and mark the feasible ones; do one pivot; name binding constraints.

Watch out: θ=0 is degeneracy, not infeasibility; the ratio test uses only decreasing variables; an unbounded region ≠ an unbounded objective.

L4 · DUALITY & SENSITIVITY

In one line: every LP has a dual that gives bounds and resource prices; sensitivity asks how the answer moves when data change.

Covered: Lagrangian relaxation (min x^2+y^2, x+y=1: g(p)=p−p^2/2, best p=1, z=½); LP dual from the Lagrangian; dual of the dual = primal; weak and strong duality; economic view (a market buys Wyndor plant hours at y=(0,1.5,1), value 36 = primal); complementary slackness; what-ifs: add a constraint, add a variable, change a RHS (shadow price + range), change a profit coefficient; Excel and Gurobi sensitivity reports.

Numbers to know: plant 3 +1 hour → z=37; +9 hours → z=42, **not** 45 (outside range). Product 1 profit 3→10: old plan alone gains ≥14; new corner (4,3), z=55 (+19).

Quiz angle: write a dual; use complementary slackness to get y*; read a shadow price and its valid range.

Watch out: shadow prices hold only inside the range and one change at a time; binding does not mean a positive price.

L5 · LP CASES: MARKDOWN & ROBUST SOURCING

In one line: two real LP cases, plus planning for the worst case when data are uncertain.

Covered: markdown management for seasonal goods (stock fixed before the season; only price can change); data → demand model → optimize; LP in "effective weeks" at each price; effect of initial stock; dynamic adjustment (re-solve with remaining stock and time). **Dual-sourcing drug case:** nominal LP; uncertain raw-material yields; ad-hoc fix; "prepare for the worst" robust LP; box and budgeted uncertainty; duality turns the worst case into linear rows.

Numbers to know: markdown optimum 11⅔ weeks at \$60 + 3⅓ at \$54, revenue \$116,750. Drug: nominal profit ≈8,819.66 can miss the ingredient target; robust ≈8,294.57 and switches to raw 1.

Quiz angle: formulate the markdown LP; write a robust row (pick the worst coefficient); contrast sensitivity vs uncertainty.

Watch out: purchase cost is sunk; with q≥0 in a ≥ row, the worst case uses the **lowest** yields.

L6 · NETWORK FLOW MODELS

In one line: many LPs are flows on a graph; that structure gives integer answers and fast special methods.

Covered: graph terms (directed/undirected, path, cycle, acyclic, connected, tree, spanning tree with n−1 arcs, bipartite, complete); min-cost flow (out − in = supply); node–arc incidence matrix; special cases: transportation, assignment, shortest path, max flow; initial solutions by northwest corner or least cost; m+n−1 basic cells form a spanning tree; transportation simplex with potentials. Applications: container shipping between Singapore, Ho Chi Minh and Bangkok (time-expanded network); expediting and outsourcing for a complex machine (project crashing); process flexibility with random demand.

Formulas: integer supplies/capacities ⇒ integer optimal flow (TU). Max flow = min cut. Complete graph n(n−1)/2 edges.

Quiz angle: write flow-balance rows; build an initial transport BFS; name the special model.

Watch out: need ∑b=0; one balance row is redundant; basic arcs can carry zero flow.

L7 · INTEGER PROGRAMMING

In one line: binary and integer variables model yes/no choices and logic; solve with LP relaxation + branch and bound.

Covered: MIP, pure IP, BIP; **if-then logic for project selection (Side 2)**; product of binaries; binary × continuous; capital budgeting; facility location; flexibility design with K demand samples; TSP (brute force, subtour cuts, MTZ); UberPOOL pickup/drop-off routing; Sudoku; Rue La La pricing (fix total price k); job scheduling; LP relaxation; Cramer's rule and totally unimodular matrices; branch and bound.

Numbers to know: knapsack LP 1150, rounded 950, IP 1000. B&B: root 2090 → optimum 1950.

Quiz angle: translate if-then rules; write big-M rows; run a few B&B nodes; explain why rounding fails.

Watch out: assignment rows alone allow subtours; use the smallest valid M.

L8 · DYNAMIC PROGRAMMING

In one line: split a sequential problem into stages, summarize the past in a state, and solve backward with the Bellman equation.

Covered: deterministic vs stochastic DP; Fibonacci (save sub-results, don't recompute); value function, Bellman equation, boundary; staged shortest path A→J; choosing the state; investment/knapsack with budget as state; Go (~250 moves per state, ~150 per game: far too many states); stochastic two-day newsvendor; open-loop vs adaptive policies; secretary (optimal stopping) with a threshold.

Numbers to know: path A→D→F→I→J = 74; knapsack 35; newsvendor profit .4 adaptive vs .35 open-loop; secretary n=4 success 11/24; large n skip ≈37%.

Quiz angle: define stage/state/action/transition; fill a DP table; write Bellman with E[·].

Watch out: take the expectation **before** the min when you act before demand is seen; infeasible states get ±∞.

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SIDE 2 / 2 · TURNING "IF… THEN…" RULES INTO MATH (BINARY LOGIC)

Set-up: six candidate projects; x_i=1 means "do project i", x_i=0 means "skip it". Every rule below becomes a **linear** constraint on these switches, so the model stays an IP the solver can handle. **Good habit:** test each rule by plugging in 0s and 1s and checking it accepts every allowed combo and rejects every forbidden one.

1 · THE CORE RULES (projects $x_1…x_6$, each a 0/1 switch)

Rule in words	Math	Why it works	0/1 test: the only combos it blocks
At most one of projects 1, 2, 3	$x_1 + x_2 + x_3 \le 1$	Only one switch can be on	any two or all three on
Exactly two of projects 2–6	$x_2 + x_3 + x_4 + x_5 + x_6 = 2$	Exactly two switches on	any count other than 2 (0, 1, 3, 4, 5 on)
If 1, then 4	$x_1 \le x_4$	If $x_1 = 1$, x_4 is forced to be 1	(x_1, x_4) = (1, 0) only
If 2, then NOT 5	$x_2 + x_5 \le 1$	They can't both be on	(x_2, x_5) = (1, 1) only
If not 1, then 6	$x_1 + x_6 \ge 1$	At least one of them must be on	(x_1, x_6) = (0, 0) only
4 and 5 go together	$x_4 = x_5$	Both on or both off	(1, 0) and (0, 1)
If 3, then 1 or 5	$x_3 \le x_1 + x_5$	Doing 3 needs at least one helper	(x_3, x_1, x_5) = (1, 0, 0) only
If 3 AND 4, then 6	$x_6 \ge x_3 + x_4 - 1$	The right side only reaches 1 when both are on	(x_3, x_4, x_6) = (1, 1, 0) only

2 · MORE RULES IN THE SAME STYLE

Rule in words	Math	Why it works / what it blocks
If 2, then 3 AND 4	$x_2 \le x_3$ and $x_2 \le x_4$	One row per required project. $2x_2 \le x_3 + x_4$ is also valid for 0/1 but gives a weaker LP relaxation
If 1 or 2, then 3	$x_1 \le x_3$ and $x_2 \le x_3$	Either trigger alone forces 3. Not $x_1 + x_2 \le x_3$ (that wrongly bans doing 1 and 2 together)
If 1 and 2, then NOT 3	$x_1 + x_2 + x_3 \le 2$	Blocks only all three on
If 1, then at least 2 of {3, 4, 5}	$x_3 + x_4 + x_5 \ge 2x_1$	$x_1 = 0$ gives ≥ 0 (no effect); $x_1 = 1$ gives ≥ 2
If 1, then none of {2, 3}	$x_2 + x_3 \le 2(1 - x_1)$	$x_1 = 1$ makes the right side 0; $x_1 = 0$ makes it 2 (no effect)
Exactly one of 1, 2 (either/or)	$x_1 + x_2 = 1$	Blocks both off and both on
1 and 2 can't both be skipped	$x_1 + x_2 \ge 1$	Same as "if not 1 then 2" and "1 unless 2"
1 only if 4 / 1 if 4	$x_1 \le x_4$ / $x_4 \le x_1$	"Only if" points at the requirement: "1 only if 4" = "if 1 then 4"
$x_1 = x_2 \cdot x_3$ (all binary)	$x_1 \le x_2$, $x_1 \le x_3$, $x_1 \ge x_2 + x_3 - 1$	$x_1 = 1$ exactly when both are on (an AND switch)
$x_1 = x_2 \cdot x_3$ (x_2 binary, $0 \le x_3 \le M$)	$-Mx_2 \le x_1 \le Mx_2$; $-M(1 - x_2) \le x_1 - x_3 \le M(1 - x_2)$	$x_2 = 0$ forces $x_1 = 0$; $x_2 = 1$ forces $x_1 = x_3$
If y = 1, then $a^Tx \le b$ (x continuous)	$a^Tx \le b + M(1 - y)$	y = 1: the real row; y = 0: M makes it slack. M = valid upper bound on $a^Tx - b$
Either $a^Tx \le b$ OR $c^Tx \le d$	$a^Tx \le b + My$; $c^Tx \le d + M(1 - y)$	y picks which row must hold; the other is switched off
Produce x > 0 only if facility y is open	$0 \le x \le My$	y = 0 forces x = 0; add fixed cost K·y to the objective

3 · METHOD, CHECKS & TRAPS

A RECIPE THAT WORKS FOR ANY RULE

1. Write it as logic: IF (conditions) THEN (results).

2. Find the one bad case: the rule is broken only when **every** condition is true and the result is false. Your constraint must cut off exactly that corner.

3. Literals: "i is done" → x_i; "i is skipped" → $(1 - x_i)$.

4. One-line formula for "THEN A OR B": ∑(1 − condition literals) + ∑(result literals) ≥ 1. For "THEN A AND B", write one row per result.

Examples: If 3 AND 4 then 6: $(1-x_3)+(1-x_4)+x_6 \ge 1 \Rightarrow \mathbf{x_6 \ge x_3+x_4-1}$. If 3 then 1 or 5: $(1-x_3)+x_1+x_5 \ge 1 \Rightarrow \mathbf{x_3 \le x_1+x_5}$. If not 1 then 6: $x_1+x_6 \ge 1$. If 2 then not 5: $(1-x_2)+(1-x_5) \ge 1 \Rightarrow \mathbf{x_2+x_5 \le 1}$.

THE 0/1 TEST, WORKED

List every 0/1 combo (2 variables = 4 rows, 3 = 8). Mark what the words allow, then plug in.

$\mathbf{x_6 \ge x_3+x_4-1}$, combos (x_3,x_4,x_6): 000 ✓ 001 ✓ 010 ✓ 011 ✓ 100 ✓ 101 ✓ **110** ✗ 111 ✓. Only "3 and 4 on, 6 off" fails. Correct.

$\mathbf{x_3 \le x_1+x_5}$, combos (x_3,x_1,x_5): only **100** gives 1 ≤ 0 ✗; all 7 others pass. Correct.

Wrong try $\mathbf{x_6 \ge x_3}$ and $\mathbf{x_6 \ge x_4}$: blocks 100-style rows (3 on, 4 off, 6 off) the words allow. That means "if 3 OR 4 then 6": too strong.

COMMON TRAPS

Direction: "if 1 then 4" is $x_1 \le x_4$. Writing $x_4 \le x_1$ means the converse "if 4 then 1".

Contrapositive is the same rule: "if not 4 then not 1" also gives $x_1 \le x_4$.

Three look-alikes: "if not 1 then 6": $x_1+x_6 \ge 1$. "If 1 then not 6": $x_1+x_6 \le 1$. "Exactly one of 1, 6": $x_1+x_6 = 1$.

OR uses a sum: "if 3 then 1 or 5" is $x_3 \le x_1+x_5$. $2x_3 \le x_1+x_5$ would demand both.

AND in the condition needs −1: $x_6 \ge x_3+x_4-1$; when only one is on the right side is 0 and nothing is forced.

An if-then never forces anything when the condition is off. Check the "condition off" rows are all still allowed.

Big-M: choose the smallest valid M; test y = 0 and y = 1 separately.

QUICK DRILL (cover the answers)

a) At least 2 of projects 1–4 → $x_1+x_2+x_3+x_4 \ge 2$

b) If 5, then neither 1 nor 6 → $x_5+x_1 \le 1$, $x_5+x_6 \le 1$

c) If not 4, then 5 and 6 → $1-x_4 \le x_5$, $1-x_4 \le x_6$

d) If 2, then exactly one of 3, 4 → $x_3+x_4 \ge x_2$, $x_3+x_4 \le 2-x_2$

e) At most 3 of projects 1–6, and if 6 then 1 → $\sum x_i \le 3$, $x_6 \le x_1$

f) Do 1 if and only if both 2 and 3 → $x_1 \le x_2$, $x_1 \le x_3$, $x_1 \ge x_2+x_3-1$